

Article

Communication Protocol Design for IoT-Enabled Energy Management in a Smart Microgrid

Shama Naz Islam ^{1,*}  and Md Apel Mahmud ²¹ School of Engineering, Deakin University, Geelong, VIC 3216, Australia² College of Science & Engineering, Flinders University, 1284 South Road Tonsley, Adelaide, SA 5042, Australia; apel.mahmud@flinders.edu.au

* Correspondence: shama.i@deakin.edu.au

Abstract: In this paper, a new communication protocol is proposed to allow direct communication between internet of things (IoT)-enabled home energy management systems (HEMSs) in a smart microgrid. The direct communication features are an important attribute for decentralised demand management and local energy trading operations in a microgrid equipped with renewable energy resources. The proposed scheme utilises the intermediate HEMSs as relay nodes that forward the sum of the received signals from nearby HEMSs to both ends of the entire network. The scheme can achieve lower latency compared to the cases when HEMSs adopt direct decode-and-forward (DF) or transmit through the central controller. For the proposed protocol, we have analytically obtained expressions for the error probability at different HEMSs, as well as the average bit error rate (BER) to indicate the overall error performance of the microgrid communication. To evaluate the proposed protocol in different channel conditions, numerical simulation is performed. The results demonstrate that the channel conditions between the HEMSs at the middle of the network have a greater impact on the system error performance. Overall, it can be observed that the proposed protocol suffers a lower degradation in error performance in comparison to direct DF when one of the users experiences worse channel conditions.

Keywords: energy management; decode-and-forward; microgrid; error performance; wireless communication



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1. Introduction

The unprecedented integration of distributed renewable energy resources (DERs) into customer premises has led to the evolution of the microgrid. Microgrids have allowed the utilisation of local energy generation to meet local energy demand with better power quality resulting from reduced transmission line losses [1–5]. Moreover, microgrids can enhance the energy security of remotely located communities by reducing dependency on the main power grid. To enable the reliable operation of a microgrid, effective energy management systems (EMSs) need to be incorporated so as to control energy sharing within the microgrid, as well as with the main power grid during contingencies [6]. For effective energy management operation, it is essential that the power generation and consumption information in the microgrid is available in real time with reasonable accuracy, hence arises the need for low-latency and high-reliability communication protocols [7–9]. In essence, this study investigates the design of a new communication scheme that enables direct communication between HEMSs for the purpose of energy management, where intermediate HEMSs relay the messages from adjacent HEMSs throughout the network of HEMSs.

With the advent of information and communication technologies, the concept of ubiquitous connectivity among every possible device that we use in our day-to-day life has received significant attention from researchers and the broader community. Such ubiquitous connectivity among all intelligent devices is often denoted as the internet of things (IoT). A number of research papers have outlined the attributes and characteristics of IoT-enabled smart cities, where IoT has applications in different sectors, ranging from agriculture, health monitoring, transport systems, financial services, as well as energy systems [10–14]. The integration of intelligent control and communication technologies in power system operation is a critical parameter to realise the envisioned future of IoT-enabled smart cities. Thus, it can be predicted that the future power system will largely depend upon smart devices and advanced technologies, leading to a smart power system network.

It goes without saying that a microgrid will need to comply with the features and requirements of a smart energy system, as microgrids are an essential component of the power system network. In a smart microgrid, the residential end-users will be equipped with intelligent home energy management systems (HEMSs), which can forward the energy generation and consumption information to a control centre or central EMS. Enabling HEMSs with IoT technology can allow for the advanced monitoring, control and energy management of a microgrid.

In a microgrid where users are equipped with renewable energy resources, there could be certain time periods when some users have excess energy generated and some other users have an energy deficit. This is not an uncommon situation since renewable energy generation is intermittent and users' energy demand can depend on lifestyle patterns or weather conditions. In such cases, local (peer-to-peer) energy trading has evolved as a natural solution to satisfy the local demand with local generation [15–18]. Local energy trading applications require energy generation/consumption information to be exchanged in real time, and hence, such applications would be much more effective when integrated with IoT technology.

Existing research on IoT-enabled smart energy systems have focussed on energy management applications and the secure operation of the power system network. The authors in [19] have investigated a building management system with IoT-enabled smart buildings and discussed the key requirements for performing a number of different functions, including demand response and building surveillance. The authors in [20] have considered the optimum scheduling and energy efficient operation of an IoT-enabled smart city which has the features of wireless power transfer. The authors have elaborated the requirements of lightweight protocols, predictive models for energy forecast and context-aware energy management for smart energy systems. IoT-enabled wind energy systems that are capable of data acquisition, supervisory control and condition monitoring have been illustrated in [21]. The aforementioned research papers have mainly reviewed the requirements and expectations of the communication protocols required for an IoT-enabled microgrid.

On the other hand, the authors in [22] have utilised the fog computing platform for energy management operations and defined the software, hardware and communication architecture for such IoT-enabled energy management systems. A communication scheme for blockchain-based peer-to-peer energy trading has been proposed by [23], where the authors investigate metadata exchange for energy transaction blocks. For real-time operation, the authors in [24] have proposed a software infrastructure to integrate the real-time energy profile and renewable generation data for scheduling applications in an IoT-enabled microgrid. This can be useful for obtaining the optimum control policies for building management. The existing research papers have mostly investigated IoT-enabled smart energy systems at a conceptual level or from an application perspective. However, the

effectiveness of IoT technology will also depend on the reliability and security features of other aspects in the communication model, e.g., the network and physical properties.

From a communication technology perspective, different wired and wireless communication schemes have been adopted in smart grid applications. Though wired communication can offer stronger reliability and robustness, wireless communication technologies have more flexibility and less dependence on infrastructure [25]. As a result, wireless technologies are emerging as a popular solution in smart energy applications in the residential context. For example, ZigBee has been deployed for appliance monitoring and sensing applications due to its low power requirement as well as low cost. They can allow for a transmission range of less than 100 m, which is suitable for residential energy applications [26]. Wireless mesh networks have also been a popular choice for smart meter deployment, where each meter has the ability to route the energy consumption data to the nearby smart meters and form a mesh [27]. Though such technologies can enhance coverage, the implementation cost and complexity can be higher. More critical energy applications such as transmission and distribution network monitoring commonly rely on cellular infrastructure, such as LTE, 4G and 5G networks [28,29]. Though they have a well-developed infrastructure that can be useful in accommodating energy applications, the infrastructure is shared with cellular applications involving voice and data usage. Another emerging technology for smart grid communications includes WiMax, which offers the appropriate designed security and backhaul connectivity mechanisms for critical applications [30]. However, the technology could not progress as expected due to a more widespread infrastructure provided by LTE and 5G networks.

Though the aforementioned technologies have been widely used for smart grid applications, they need to be adapted to meet the requirements for IoT-enabled smart grid components. IoT applications heavily rely on low power wide area network (LPWAN) technologies, as it allows IoT devices to operate at a lower power (thus improving the battery life) and communicate over a wider area. It should be noted that in the context of local energy trading, the energy supply/demand balance is achieved within nearby households alone, which are usually within a few kilometres from each other. In this regard, the range offered by LPWAN technologies is appropriate for local energy trading applications. There are a number of LPWAN technologies that are currently being deployed in IoT applications. Some of these protocols operate over an unlicensed spectrum (e.g., LoRa and Sigfox), while others operate over the licensed band (e.g., Long-Term Evolution for Machine type communication (LTE-M) and Narrow Band IoT (NB-IoT)) [31]. Existing research in this domain has focussed on bit error rate (BER) analysis of LoRa technology [32], as well as latency, collision rate and throughput analysis [33].

Though LoRa and Sigfox protocols are widely used by IoT devices, the protocols suffer from efficiency issues due to collisions with interference signals and the resulting packet losses [34]. The authors in [35] have demonstrated that the coverage probability for LoRa networks decreases exponentially with an increase in the number of end devices due to the interference issue. Similar outcomes have also been presented in [36] for LoRa and Sigfox protocols through experimental setups in northern Denmark, where the authors reported a significant increase in failure rate due to external interference. To counter the interference problem, retransmissions for multiple times are usually adopted as a solution [37], which is often not suitable for smart energy applications that require low latency and high reliability. Moreover, this can lead to a trade-off between energy consumption and reliability depending on the number of retransmissions [34]. An alternative strategy could be the use of precoding methods to maximise the transmission power of the legitimate users [38]. LoRa-based transmissions have been used in [39] to design the communication between smart meters and microcontrollers for a partially grid-connected microgrid to demonstrate

the suitability of LoRa technologies in microgrid applications. The use of the unlicensed spectrum in LoRa and Sigfox technologies can expose the smart grid communication system to security threats and additional security mechanisms may need to be incorporated before their application in the energy system context.

The NB-IoT protocol utilises the LTE band or re-farmed GSM band, though it adopts repetition mechanism to enhance coverage with low complexity [40]. To reduce collision at the preamble, an access reservation protocol is proposed in [41] that is based on partial preamble transmission. The authors in [42] have investigated the uplink coverage for NB-IoT and obtained that coverage significantly depends on the channel estimation and coherence time. To realise the actual characteristics of NB-IoT systems, the authors have developed a simulation model for NB-IoT uplink and downlink [43]. Regarding smart grid applications, the authors in [44] have examined different IoT protocols for different smart grid applications and evaluated their latency. To improve the communication efficiency for IoT-enabled demand management applications, the authors in [45] have adopted an adaptive data compression mechanism and reduced the information flow in the smart metering communication infrastructure.

Existing IoT communication technologies such as NB-IoT and LoRa have not incorporated multicast or inter-terminal communication yet. Usually, these protocols allow user equipments or IoT terminals to send messages only to the IoT base stations or gateways using separate time slots [31]. Thus, multiple IoT terminals are not allowed to transmit at the same time. However, allowing for the direct communication between IoT terminals can be useful in applications like decentralised energy management. For local energy trading operations, HEMSs can act like IoT terminals and the control centre, equipped with central EMS, can act like IoT gateways. If the generation/demand information is available only at the central EMS for optimising energy sharing among participants in a local energy market, the energy trading will be adversely affected in the case of a single point of failure at the control centre. Thus, decentralised energy management is often recommended by researchers to enhance the effectiveness and reliability of energy trading operations [46–48].

For accommodating the specific requirements for smart microgrid applications like local energy trading in a decentralised manner, a new communication protocol needs to be added to the existing IoT standards, which can support direct communication between IoT-enabled HEMSs. To address this open research problem, in this paper, we propose a communication protocol for including the feature of direct communication between IoT-enabled HEMSs. The protocol utilises the intermediate HEMSs as relay nodes which can forward the sum of the signals received from nearby HEMSs to the other HEMSs, and thus enable HEMSs to extract each other's messages without going through the central controller. The most beneficial attribute of this scheme is that it allows information exchange in a smart microgrid with a lower latency compared to the cases where decode–and–forward (DF) is employed directly or where the messages are exchanged through the central controller. To the best of our knowledge, direct communication between IoT terminals for decentralised energy management in a smart microgrid has not been addressed in the existing literature.

To be specific, we make the following contributions in this paper:

- We develop a new communication protocol for direct communication between IoT-enabled HEMSs. For the proposed protocol, we have analytically derived the mathematical expressions for error probability at a certain HEMS when decoding a signal from another HEMS. Our analysis shows that the error probabilities accumulate as the messages are forwarded across multiple HEMSs. However, each HEMS can still decode nearby HEMSs with better reliability.
- For a smart microgrid with multiple HEMSs, we have obtained mathematical expressions for the average BER in the overall network. It has been shown that the channel

conditions between the HEMSs at the middle of the network have a greater impact on the worst case error performance, as well as the average BER for the proposed protocol.

- We have performed a numerical simulation of the proposed protocol and have demonstrated the relative impact of channel conditions between different HEMSs in comparison with the benchmark direct DF mechanism. It can be observed that when one of the HEMSs suffers from poor channel conditions, the proposed protocol can still maintain the same level of error performance as compared with the case of good channel conditions at all HEMSs. However, direct DF exhibits a worse error performance even if one of the HEMSs suffers poor channel conditions.

The rest of the paper is organised in the following manner. The system model of an IoT-enabled smart microgrid integrated with direct communication features between HEMSs is presented in Section 2. The information exchange protocols are explained in Section 3 for different possible combinations of HEMS indices across different time slots. The error performance analysis for the communication between two HEMSs, as well as for the overall microgrid comprised of multiple HEMSs, is performed in Section 4. The numerical simulation results are presented in Section 5 to illustrate the effectiveness of the proposed scheme. Finally, Section 6 concludes the paper.

2. System Model

We consider a smart microgrid, composed of K number of houses equipped with solar panels, smart switches and HEMSs. The HEMSs include the functionality of scheduling the household appliances based on renewable generation in the microgrid. Given that a household may have excess generation or an energy deficit at certain points of time, they can exchange the energy among each other through local energy trading. To enable energy sharing, these HEMSs communicate with each other to identify the difference between supply and demand in each of the houses.

The effectiveness of energy trading operations largely depends on how accurately the generation and consumption information can be decoded by the HEMSs. The following example will show how the errors in the communication channel can lead to a failure in energy trading operation.

Motivating Example:

Here, we present an example of a smart microgrid with five houses equipped with HEMSs. Let us consider that House 1 and House 2 have an excess energy of 0.3 kW and 0.7 kW, respectively, at a certain time point. On the other hand, Houses 3, 4 and 5 are in a deficit of 0.15 kW, 0.65 kW and 0.1 kW, respectively, at the same time point. With no errors in the communication channel in an ideal scenario, the EMS optimises energy trading decisions as 0.15 kW power from House 1 to House 3, 0.65 kW power from House 2 to House 4 and 0.1 kW power from either House 1 or 2 to House 5. The remaining excess energy will be sold to the power utility. Now, when the communication channel has errors, the EMS obtains an inaccurate estimation of excess energy from Houses 1 and 2 as 0.2 kW and 0.35 kW, respectively. On the other hand, there are also errors in the estimated demand values for Houses 3, 4 and 5 as 0.2 kW, 0.45 kW and 0.05 kW, respectively. Now, the EMS will make non-optimal trading decisions, delivering 0.2 kW from House 1 to House 3 and 0.35 kW from House 2 to House 4, thus leaving House 4 and House 5 in a deficit of 0.3 kW and 0.1 kW, respectively, which the houses will need to draw from the main power grid. However, Houses 1 and 2 had some excess energy which has to be supplied to the main grid now. Thus, the energy trading operation is not effective.

Thus, we need to develop an efficient and reliable communication scheme, which can manage communication channel errors more effectively, by adding multiple communication channels. In the aforementioned microgrid, we consider that the HEMSs coordinate the

energy sharing operation in a distributed manner. This is due to the fact that incorporating a central controller will need a huge amount of information to be exchanged between the HEMSs and the central controller. Moreover, there remains the risk of single-point failure in such a system.

However, enabling HEMSs to directly communicate with each other will allow the HEMSs to retrieve some information about each other without waiting for transmission from the central controller or gateways in IoT systems. As a result, in this paper, we consider the utilisation of overheard information from nearby HEMSs to improve information exchange reliability. That is, the IoT-enabled HEMSs can act as communication relay nodes that can forward overheard information to other HEMSs and can hence create multiple communication pathways. The aforementioned system model is illustrated in Figure 1.

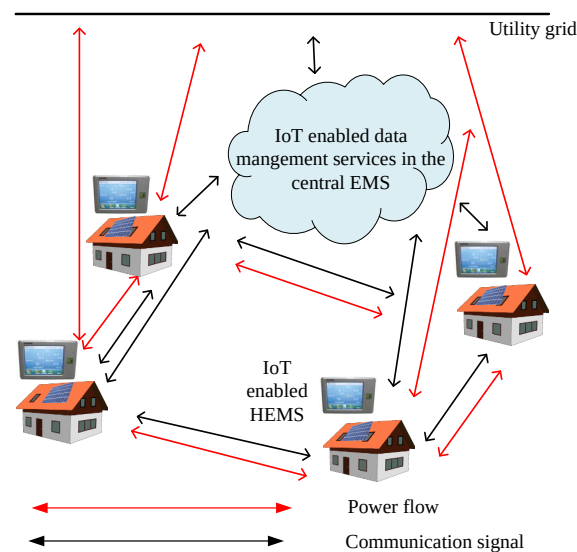


Figure 1. A microgrid comprising households with renewable generation and IoT-enabled HEMSs that can exchange excess demand/generation information with each other for local energy trading applications.

The new scheme enables all these HEMSs in the microgrid to communicate with each other without involving the central controller. At the beginning, users with odd-indexed HEMSs will transmit, whereas, the even-indexed HEMSs will act as relay nodes. Then, the even-indexed and odd-indexed HEMSs will switch their roles between transmitters and relays. The relay HEMSs receive the sum of signals from neighbouring HEMSs and forward to these HEMSs so that they can cancel self-interference to extract new signals from other HEMSs. If a HEMS receives signals from two relay HEMSs, it cancels out the common information between these two signals and extracts the new messages. Whenever a HEMS receives information about two different HEMSs, it computes the sum of the signals and forwards this sum to the nearby HEMSs acting as relays. The process repeats itself as long as all HEMSs receive each other's information.

Let us elaborate on the above communication scheme with an example system of five HEMSs. We denote the HEMSs as HEMS A, B, C, D and E. Let us consider that, initially, HEMSs A, C and E transmit the energy mismatch signals and HEMSs B and D listen. HEMS B receives the sum of signals from HEMSs A and C, whereas HEMS D receives signals from HEMSs C and E (time slot 1). Next, HEMSs B and D transmit and HEMSs A, C and E listen (time slot 2). HEMS A and E receive from HEMSs B and D, respectively, whereas HEMS C receives the sum of signals from HEMSs B and D. Then, HEMS B acts as a relay and forwards the sum of signals received in time slot 1 to A and C (time slot 3). Thus, HEMS A and HEMS C receive the messages from each other. Then, HEMS D forwards the sum

of signals to HEMSs C and E, enabling HEMSs C and E to retrieve each other's messages (time slot 4). Next, HEMS C forwards the sum of signals received in time slot 2 to HEMS B and HEMS D (time slot 5). As a result, HEMSs B and D receive messages from each other. Then, HEMS C forwards its message to HEMS B and HEMS D (time slot 6), which allows HEMSs B and D to decode the signals from HEMS A and HEMS E, respectively. Next HEMS B forwards the sum of HEMS C and D's signals, whereas HEMS D forwards the sum of HEMS B and HEMS C's signals (time slot 7). This enables HEMS A to decode HEMS D and HEMS E to decode HEMS B. Finally, HEMS C forwards the sum of signals from HEMS A and E, and HEMS B and D received this sum and decode HEMS E and HEMS A, respectively (time slot 8). Then, HEMS B (D) forwards the signal from HEMS E (A) to HEMS A (E). (time slot 9). Thus, nine time slots are needed in this scheme for the information exchange between five HEMSs.

Remark 1. *If the IoT-enabled HEMSs communicate with each other through the control centre, then, in the example system, HEMSs A, B, C, D and E will transmit their messages using five different time slots in the uplink. Then, the control centre will broadcast the message of each HEMS to all the HEMSs using five time slots in the downlink. Thus, in this scenario, a total of 10 time slots will be needed for centralised systems, similar to the ones with the NB-IoT protocol. As a result, the proposed scheme can significantly reduce the time delay and allow for better spectral efficiency compared to centralised systems.*

Remark 2. *If the intermediate nodes decode the messages directly and forward them rather than their sum, then, in the aforementioned example, HEMS B will directly receive the message from HEMS A. Then, the process will continue from B to C, C to D and D to E. Thus, after four time slots, all HEMSs will have a copy of HEMS A's message. As a result, the information sharing among all HEMSs will be completed in $5 \times 4 = 20$ time slots. In effect, significant time savings can be achieved through the proposed method compared to systems that use one-hop direct communication (such as wireless mesh networks) where all HEMSs need to know each other's messages for decentralised energy management.*

Let us consider that the set of HEMSs are denoted by $\mathcal{K} = \{1, 2, \dots, K\}$ and individual HEMSs are indexed by k . The renewable energy generation and electricity demand managed by the k th HEMS are represented by G_k and D_k , respectively. Thus, the mismatch between generation and demand at a certain time point is computed as $D_k - G_k$ at the k th HEMS. The information is then quantised, coded and modulated as u_k . The channel coefficient between the two HEMSs indexed by k and ℓ (where $\ell \in \mathcal{K}, \ell \neq k$) is denoted by $\mathbf{h}_{k,\ell}$.

The following assumptions are made about the channel coefficients:

1. The channels are considered constant within a certain time slot. The channel coefficients in one time slot are considered independent from that in another time slot.
2. All HEMSs know about their local channel state information (CSI) alone and do not have access to the global CSI.
3. The channels are modelled as Rayleigh fading channels where the channel coefficients are considered as complex-valued Gaussian random variables with zero mean and covariance $M_{i,j}$, where $i, j \in \mathcal{K}$.

The aforementioned assumptions are widely applied in smart grid communication research [44,49,50]. It should be noted that the analysis in this paper can also be applied for other channel fading models used in wireless communication research. In addition, the impact of channel estimation errors can be modelled using the approach in [51] and is expected to add a constant SNR penalty in the achievable error performance.

3. Information Exchange between HEMSs

The mathematical formulations presented in this section elaborate on the information exchange between the k th and ℓ th HEMSs at a certain time point. To be specific, we have considered the cases for k, ℓ both odd; k odd, ℓ even; k even, ℓ odd; and k, ℓ both even.

3.1. k Odd, ℓ Odd

In this case, at the first time slot, the $\bar{k}$ th HEM, where $\bar{k} \in \mathcal{K}, k \leq \bar{k} \leq \ell$ and $\bar{k} \bmod 2 \neq 0$, transmits and the $\bar{\ell}$ th HEMS, where $\bar{\ell} \in \mathcal{K}, k \leq \bar{\ell} \leq \ell$ and $\bar{\ell} \bmod 2 = 0$ receives the messages. That is, the $\bar{\ell}$ th HEMS receives the following:

$$v_{\bar{\ell}} = \begin{cases} h_{\bar{\ell}-1, \bar{\ell}} u_{\bar{\ell}-1} + n_{\bar{\ell}} & \text{if } \bar{\ell} = K \\ h_{\bar{\ell}-1, \bar{\ell}} u_{\bar{\ell}-1} + h_{\bar{\ell}+1, \bar{\ell}} u_{\bar{\ell}+1} + n_{\bar{\ell}} & \text{if } \bar{\ell} \neq K \end{cases} \quad (1)$$

The signals are decoded as $\hat{u}_{\bar{\ell}-1}$ and $\hat{u}_{\bar{\ell}+1}$ at the $\bar{\ell}$ th HEMS using the maximum likelihood (ML) algorithm.

During the second time slot, the $\bar{\ell}$ th HEMS adds up the decoded signals from the previous time slot and forwards the information to the $\bar{k}$ th HEMS. The received signal at the $\bar{k}$ th HEMS is as follows:

$$v_{\bar{k}} = \begin{cases} h_{\bar{k}+1, \bar{k}} (\hat{u}_{\bar{k}} + \hat{u}_{\bar{k}+2}) + n_{\bar{k}} & \text{if } \bar{k} = 1 \\ h_{\bar{k}-1, \bar{k}} (\hat{u}_{\bar{k}-2} + \hat{u}_{\bar{k}}) + h_{\bar{k}+1, \bar{k}} (\hat{u}_{\bar{k}} + \hat{u}_{\bar{k}+2}) + n_{\bar{k}} & \text{if } \bar{k} \neq 1 \end{cases} \quad (2)$$

The $\bar{k}$ th HEMS then decodes $\hat{V}_{\bar{k}, \bar{k}+2}$ and $\hat{V}_{\bar{k}, \bar{k}-2}$ as an estimate of $\hat{u}_{\bar{k}} \oplus \hat{u}_{\bar{k}+2}$ and $\hat{u}_{\bar{k}} \oplus \hat{u}_{\bar{k}-2}$, respectively, using ML decoding. Then, the $\bar{k}$ th HEMS performs the XOR operation between its own message and the estimates $\hat{V}_{\bar{k}, \bar{k}+2}$ and $\hat{V}_{\bar{k}, \bar{k}-2}$ to extract the signals from the $(\bar{k}-2)$ th and $(\bar{k}+2)$ th HEMSs, respectively.

Next, the signals from $(\bar{k}-2)$ th and $(\bar{k}+2)$ th HEMSs are combined at the $\bar{k}$ th HEMS who then relays the sum of the signals to the $(\bar{k}-1)$ th and $(\bar{k}+1)$ th HEMSs. The receiving HEMSs perform the XOR operation between the known and received information and obtain the unknown information. The HEMSs continue the process until the $(\frac{|\ell-k|}{2} + 1)$ th HEMS extracts the messages from the k th and ℓ th HEMSs, computes the sum and forwards it to the $(\frac{|\ell-k|}{2})$ th HEMS and $(\frac{|\ell-k|}{2} + 2)$ th HEMS. It takes up to $\frac{|\ell-k|}{2}$ time slots to achieve this. Note that, $|\ell-k|$ is an even number when both ℓ and k are odd numbers.

Finally, the $(\frac{|\ell-k|}{2})$ th HEMS receives the signal, as follows:

$$v_{\frac{|\ell-k|}{2}} = h_{\frac{|\ell-k|}{2}+1, \frac{|\ell-k|}{2}} (\hat{u}_k + \hat{u}_\ell) + n_{\frac{|\ell-k|}{2}} \quad (3)$$

Similarly, the $(\frac{|\ell-k|}{2} + 2)$ th HEMS receives the signal as follows:

$$v_{\frac{|\ell-k|}{2}+2} = h_{\frac{|\ell-k|}{2}+3, \frac{|\ell-k|}{2}+2} (\hat{u}_{k+1} + \hat{u}_{\ell-1}) + n_{\frac{|\ell-k|}{2}+2} \quad (4)$$

Both the $(\frac{|\ell-k|}{2})$ th and $(\frac{|\ell-k|}{2} + 2)$ th HEMSs extract the XOR of the messages from the ℓ th and k th HEMSs. The extracted messages are forwarded to the nearby HEMSs, who then continue to forward the sum of the messages to the adjacent HEMSs until the k th and the ℓ th HEMSs receive the messages from the $(k+1)$ th and $(\ell-1)$ th HEMSs, respectively. Finally, the k th and ℓ th HEMSs cancel their own signal to extract each other's messages.

We explain the above mechanism with the following example:

Example: Let us consider a microgrid with seven HEMSs, A, B, C, D, E, F, G, where HEMS A wants to exchange information with HEMS G at a certain instant (to model the worst case scenario). At the first time slot, HEMSs A, C, E and G transmit and HEMSs B, D and F receive the sum of signals from HEMS A and C, HEMS C and E and HEMS E

and G, respectively. During the second time slot, HEMSs B, D and F decode the signals from HEMS A and C, HEMS C and E and HEMS E and G before forwarding the sum to the corresponding HEMSs, respectively. Then, HEMS A and C, C and E and E and G extract each other's signals. In the third time slot, HEMS C computes the sum of the signals from HEMSs A and E and forwards that to HEMS D. At the same time, HEMS E forwards the sum of signals from HEMSs C and G to HEMS D. HEMS D already knows the signals from HEMSs C and E (from the first time slot). Thus, it cancels the known information and computes the sum of signals from HEMSs A and G. In the fourth time slot, HEMS D forwards the sum of the signals from HEMSs A and G to HEMS C and E. During time slot 5, HEMSs C and E forwards the signal to HEMSs B and F, respectively. Finally, in time slot 6, HEMSs B and F forward the signal to HEMS A and G, respectively. After this, HEMSs A and G cancel their own signals and extract each other's messages.

3.2. k Odd, ℓ Even

In this case, during the first time slot, the $\bar{k}$ th HEMS ($\bar{k} \in \mathcal{K}, k \leq \bar{k} \leq \ell$ and $\bar{k} \bmod 2 \neq 0$) transmit and the $\bar{\ell}$ th HEMS ($\bar{\ell} \in \mathcal{K}, k \leq \bar{\ell} \leq \ell$ and $\bar{\ell} \bmod 2 = 0$) receives the messages, as in (1), and estimates them as $\hat{u}_{\bar{\ell}-1}$ and $\hat{u}_{\bar{\ell}+1}$.

On the other hand, in the second time slot, the $\bar{\ell}$ th HEMS ($\bar{\ell} \in \mathcal{K}, k \leq \bar{\ell} \leq \ell$ and $\bar{\ell} \bmod 2 = 0$) transmits and the $\bar{k}$ th HEMS ($k \in \mathcal{K}, k \leq \bar{k} \leq \ell$ and $\bar{k} \bmod 2 \neq 0$) receives as follows:

$$v_{\bar{k}} = \begin{cases} h_{\bar{k}-1, \bar{k}} u_{\bar{k}-1} + n_{\bar{k}} & \text{if } \bar{k} = 1 \\ h_{\bar{k}-1, \bar{k}} u_{\bar{k}-1} + h_{\bar{k}+1, \bar{k}} u_{\bar{k}+1} + n_{\bar{k}} & \text{if } \bar{k} \neq 1 \end{cases} \quad (5)$$

The received signals are estimated as $\hat{u}_{\bar{k}-1}$ and $\hat{u}_{\bar{k}+1}$.

In the third time slot, the $\bar{\ell}$ th HEMS forwards the sum of the decoded signals from the first time slot to the $\bar{k}$ th HEMS, where the received sum of signals is given by (2). The $\bar{k}$ th HEMS then decodes $\hat{V}_{\bar{k}, \bar{k}+2}$ and $\hat{V}_{\bar{k}, \bar{k}-2}$ as an estimate of the corresponding sum of the signals. After this, the $\bar{k}$ th HEMS computes $\hat{V}_{\bar{k}, \bar{k}+2} \oplus u_{\bar{k}}$ and $\hat{V}_{\bar{k}, \bar{k}-2} \oplus u_{\bar{k}}$ to obtain the signals from the $(\bar{k} - 2)$ th and $(\bar{k} + 2)$ th HEMSs.

Similarly, during time slot 4, the HEMSs indexed by $\bar{k}$ compute the sum of the decoded signals from time slot 2 and forward that to the HEMSs indexed by $\bar{\ell}$. Thus, the $\bar{\ell}$ th HEMSs receive the signal as follows:

$$v_{\bar{\ell}} = \begin{cases} h_{\bar{\ell}+1, \bar{\ell}} (\hat{u}_{\bar{\ell}} + \hat{u}_{\bar{\ell}+2}) + n_{\bar{\ell}} & \text{if } \bar{\ell} = K \\ h_{\bar{\ell}-1, \bar{\ell}} (\hat{u}_{\bar{\ell}-2} + \hat{u}_{\bar{\ell}}) + h_{\bar{\ell}+1, \bar{\ell}} (\hat{u}_{\bar{\ell}} + \hat{u}_{\bar{\ell}+2}) + n_{\bar{\ell}} & \text{if } \bar{\ell} \neq K \end{cases} \quad (6)$$

The $\bar{\ell}$ th HEMS then decodes $\hat{V}_{\bar{\ell}, \bar{\ell}+2}$ and $\hat{V}_{\bar{\ell}, \bar{\ell}-2}$ as an estimate of $\hat{u}_{\ell} \oplus \hat{x}_{\ell+2}$ and $\hat{u}_{\ell} \oplus \hat{x}_{\ell-2}$, respectively. After this, the ℓ th HEMS cancels its own message to obtain the $(\ell - 2)$ th and $(\ell + 2)$ th HEMSs' signals.

The process continues until the $(\frac{|\ell-k|+1}{2} + 1)$ th time slot. After that, the $(\frac{|\ell-k|+1}{2})$ th HEMS obtains the signal of the k th and $(\ell - 1)$ th HEMSs. On the other hand, the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS obtains the signal of the $(k + 1)$ th and ℓ th HEMSs. Then, during the $(\frac{|\ell-k|+1}{2} + 2)$ th time slot, the $(\frac{|\ell-k|+1}{2})$ th HEMS forwards the sum of the received signals to the two adjacent HEMSs. In effect, the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS now receives

$$v_{\frac{|\ell-k|+1}{2}+1} = h_{\frac{|\ell-k|+1}{2}+1, \frac{|\ell-k|+1}{2}} (\hat{u}_k + \hat{u}_{\ell-1}) + n_{\frac{|\ell-k|+1}{2}+1} \quad (7)$$

Similarly, during the $(\frac{|\ell-k|+1}{2} + 3)$ th time slot, the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS forwards the sum of the signals and the $(\frac{|\ell-k|+1}{2})$ th HEMS receives

$$v_{\frac{|\ell-k|+1}{2}} = h_{\frac{|\ell-k|+1}{2}, \frac{|\ell-k|+1}{2}+1} (\hat{u}_{k+1} + \hat{u}_{\ell}) + n_{\frac{|\ell-k|+1}{2}} \quad (8)$$

After these two time slots, the $(\frac{|\ell-k|+1}{2})$ th HEMS decodes $\hat{V}_{k+1,\ell}$ as an estimate of $\hat{u}_{k+1} \oplus \hat{u}_\ell$. Since the $(\frac{|\ell-k|+1}{2})$ th HEMS already knows $(k+1)$ th HEMS's signal, it can cancel the known information to extract the ℓ th HEMS's signal. Similarly, the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS can decode $\hat{V}_{k,\ell-1}$ and then utilise the $(\ell-1)$ th HEMS's signal to obtain the message from the k th HEMS.

Once the $(\frac{|\ell-k|+1}{2})$ th HEMS knows the message from the ℓ th HEMS, it computes the sum of signals of the k th and ℓ th HEMSs and then forwards this message in the subsequent time slot to HEMS $\frac{|\ell-k|+1}{2} - 1$. Similarly, the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS relays the superposition of the k th and ℓ th HEMSs' signals to HEMS $(\frac{|\ell-k|+1}{2}) + 2$. The $(\frac{|\ell-k|+1}{2} - 1)$ th and $(\frac{|\ell-k|+1}{2} + 2)$ th HEMSs decode the sum of the signals as $\hat{V}_{k,\ell}$ and then forward these to the nearby HEMSs. The message exchange mechanism keeps repeating until the k th and the ℓ th HEMS have received the messages from the $(k+1)$ th and the $(\ell-1)$ th HEMSs, respectively. Finally, the k th and the ℓ th HEMSs compute $\hat{V}_{k,\ell} \oplus u_k$ and $\hat{V}_{k,\ell} \oplus u_\ell$, respectively to obtain the ℓ th and k th HEMSs' messages.

The following example can illustrate the aforementioned process:

Example: Let us consider the example microgrid from Section 3.1. Now, HEMS A wants to exchange the information with HEMS F at a certain instant. At the first time slot, HEMSs A, C, E and G transmit and HEMSs B, D and F receive the sum of signals from HEMS A and C, HEMS C and E and HEMS E and G, respectively. During the second time slot, HEMSs B, D and F transmit their messages, whereas HEMSs C and E receive the sum of signals from HEMSs B and D and HEMSs D and F, respectively. On the other hand, HEMSs A and G receive from HEMS B and HEMS F, respectively. Then in time slot 3, HEMSs B, D and F decode the signals from HEMS A and C, HEMS C and E and HEMS E and G, before forwarding the sum to the corresponding HEMSs, respectively. Then, HEMS A and C, C and E and E and G extract each other's signals. In the fourth time slot, HEMSs C and E decode the signals from HEMS B and D and HEMS D and F, respectively, and forward the sum to the corresponding HEMSs. Then, HEMSs B and D and D and F extract each other's signals. Also, HEMSs A and G decode the signals from HEMS B and F. After this, HEMS C extracts the signal from HEMSs A and E and forwards the sum to HEMS D in time slot 5. On the other hand, HEMS D extracts the signals from HEMSs B and F and forwards the sum to HEMS C in time slot 6. HEMS D already knows the signal from HEMS E (from time slot 2). Thus, it cancels the known information and computes the signal from HEMS A. Similarly, HEMS C already knows the signal from HEMS B (from time slot 2) and cancels this to extract the signal from HEMS F. Then in the seventh time slot, HEMSs C and D compute the sum of the signals from HEMSs A and F and forward that to HEMSs B and E. During time slot 8, HEMSs B and E forward the signal to HEMSs A and F, respectively. After this, HEMSs A and F cancel their own signals and extract each other's messages.

3.3. k Even, ℓ Odd

The signal transmission will be similar to that in Section 3.2. Instead of the odd-indexed HEMSs, the signal exchange will start with the transmission from the even-indexed HEMSs. So, the signals in (5) and (6) will be applicable to this case, with $\bar{k}$ replaced by $\bar{\ell}$ and vice versa. After the $(\frac{|\ell-k|+1}{2} + 1)$ th time slot, the $(\frac{|\ell-k|+1}{2} + 2)$ th and the $(\frac{|\ell-k|+1}{2} + 3)$ th HEMS exchange messages with each other to retrieve the messages from the ℓ th and k th HEMSs. Thus, (7) and (8) can be applied to this case by replacing $\frac{|\ell-k|+1}{2}$ by $\frac{|\ell-k|+1}{2} + 2$ and replacing $\frac{|\ell-k|+2}{2}$ by $\frac{|\ell-k|+3}{2}$. The remaining process is similar to the one in Section 3.2.

3.4. k Even, ℓ Even

The signal transmission will be similar to the one in Section 3.1, with $\bar{k}$ replaced by $\bar{\ell}$. That is, the transmissions will begin with the even-indexed HEMSs, and the odd-indexed HEMSs will act as relays.

4. Error Probability Analysis

In this section, we have analytically investigated the error performance when the k th HEMS obtains the signal from the ℓ th HEMS. Similarly to the previous section, we'll consider the cases k odd, ℓ odd and k odd, ℓ even. The case k even, ℓ odd will have similar expressions for error performance as the second case (k odd, ℓ even). On the other hand, case k even, ℓ even will yield similar results to the first case (k odd, ℓ odd). This analysis is based on binary phase shift keying (BPSK) modulation at the HEMSs, but this can be similarly extended to quadrature phase shift keying (QPSK) modulation, since QPSK transmissions can be considered as two orthogonal BPSK transmissions.

4.1. k Odd, ℓ Odd

In this case, during the first time slot, the probability of error at the ℓ th HEMS is given by the following lemma.

Lemma 1. *The error probability at the $\bar{\ell}$ th HEMS (where $\bar{\ell}$ is even) during the first time slot is as follows:*

$$Pr_{\bar{\ell}} = \begin{cases} Q\left(\sqrt{\frac{2|h_{\bar{\ell}-1,\bar{\ell}}|^2}{N_0}}\right) & \text{if } \bar{\ell} = L \\ Q\left(\sqrt{\frac{2|h_{\bar{\ell}-1,\bar{\ell}}|^2}{N_0}}\right)(1 - Q\left(\sqrt{\frac{2|h_{\bar{\ell}-1,\bar{\ell}}|^2}{N_0}}\right)) + Q\left(\sqrt{\frac{2|h_{\bar{\ell},\bar{\ell}+1}|^2}{N_0}}\right)(1 - Q\left(\sqrt{\frac{2|h_{\bar{\ell}-1,\bar{\ell}}|^2}{N_0}}\right)) & \text{if } \bar{\ell} \neq L \end{cases} \quad (9)$$

Proof. Note that at the first time slot, the $\bar{\ell}$ th HEMS decodes the message from the $(\bar{\ell} - 1)$ th HEMS if $\bar{\ell} = L$ or the sum of signals from the HEMSs $\bar{\ell} + 1$ and $\bar{\ell} - 1$ if $\bar{\ell} \neq L$. The SNR of the signal received from the $(\bar{\ell} - 1)$ th HEMS at the $\bar{\ell}$ th HEMS is given by $\frac{|h_{\bar{\ell}-1,\bar{\ell}}|^2}{N_0}$. If BPSK modulation is considered, the probability of incorrectly decoding the message from the $(\bar{\ell} - 1)$ th HEMS at the $\bar{\ell}$ th HEMS is written as in the first expression in (9).

When $\bar{\ell} \neq L$, the $\bar{\ell}$ th HEMS can incorrectly decode either of the signals from the $(\bar{\ell} - 1)$ th HEMS or the $(\bar{\ell} + 1)$ th HEMS to incorrectly decode the sum of signals from the $(\bar{\ell} + 1)$ th and $(\bar{\ell} - 1)$ th HEMSs. In case the $\bar{\ell}$ th HEMS incorrectly decodes both the other HEMSs' signals, then it can still decode the signal correctly due to the signal patterns in the BPSK modulation. As a result, the probability of incorrectly decoding the sum of the signals from the $(\bar{\ell} - 1)$ th and the $(\bar{\ell} + 1)$ th HEMSs at the $\bar{\ell}$ th HEMS is given as in the second expression in (9), where the first (second) term indicates the probability of incorrectly decoding the signal from the $(\bar{\ell} - 1)$ th ($(\bar{\ell} + 1)$ th) HEMS, while correctly decoding the signal from the $(\bar{\ell} + 1)$ th ($(\bar{\ell} - 1)$ th) HEMSs. $\square$

In the next time slot, if $\bar{k} = 1$, then the sum of the signals from the $\bar{k}$ th and the $(\bar{k} + 2)$ th HEMSs is decoded by the $\bar{k}$ th HEMS. In the case of $\bar{k} \neq 1$, the $\bar{k}$ th HEMS decodes the sum of signals from the HEMSs indexed by $\bar{k}$ and the $(\bar{k} - 2)$ in addition. Thus, the error probability at the $\bar{k}$ th HEMS for decoding the sum of signals from the HEMSs indexed by

$\bar{k}$ and $\bar{k} + 2$, as well as the sum of the signals from the $\bar{k}$ th and the $(\bar{k} - 2)$ th HEMS in the second time slot, is given as follows:

$$\Pr_{\bar{k}} = \begin{cases} Q(\sqrt{\frac{2|h_{\bar{k}+1,\bar{k}}|^2}{N_0}}) & \text{if } \bar{k} = 1 \\ Q(\sqrt{\frac{2|h_{\bar{k}-1,\bar{k}}|^2}{N_0}}) & \text{if } \bar{k} \neq 1 \end{cases} \quad (10)$$

Note that the expressions in (10) indicate the probability of incorrectly estimating $\hat{V}_{\bar{k},\bar{k}+2}$ and $\hat{V}_{\bar{k},\bar{k}-2}$. The expression for this probability has been derived in [52,53] and can be readily applied to the HEMSs receiving the sum of the signals.

Now, after the second time slot, the $\bar{k}$ th HEMS can extract the signal from the $(\bar{k} + 2)$ th and the $(\bar{k} - 2)$ th HEMS by cancelling self information. Thus, the probability of incorrectly decoding the new information after time slot 2 is given as in the following lemma.

Lemma 2. *The probability that the $\bar{k}$ th HEMS incorrectly decodes the signal from the $(\bar{k} + 2)$ th and the $(\bar{k} - 2)$ th HEMSs is as follows:*

$$\begin{aligned} \Pr_{\bar{k},\bar{k}+2} &= \Pr_{\bar{k}}(1 - \Pr_{\bar{k}+1}) + (1 - \Pr_{\bar{k}})\Pr_{\bar{k}+1} \\ \Pr_{\bar{k},\bar{k}-2} &= \Pr_{\bar{k}}(1 - \Pr_{\bar{k}-1}) + (1 - \Pr_{\bar{k}})\Pr_{\bar{k}-1} \end{aligned} \quad (11)$$

where $\Pr_{\bar{k}+1}$ and $\Pr_{\bar{k}-1}$ denote the probability of incorrectly decoding the signal at the $(\bar{k} + 1)$ th and the $(\bar{k} - 1)$ th HEMSs during time slot 1.

Proof. There are two criteria which determine whether the signal of the $(\bar{k} + 2)$ th HEMS can be obtained accurately. These include the following: (i) what $(\bar{k} + 1)$ th HEMS decodes in the first time slot, and (ii) what $\bar{k}$ th HEMS decodes in the second time slot. Thus, the $\bar{k}$ th HEMS can erroneously detect the $(\bar{k} + 2)$ th HEMS's signal in two possible ways as follows: (i) if $(\bar{k} + 1)$ th HEMS decodes $\hat{u}_k + \hat{u}_{k+2}$ incorrectly and at the same time, $\hat{V}_{k,k-2}$, forwarded from the $(\bar{k} - 1)$ th HEMS, is correctly decoded at the $\bar{k}$ th HEMS, (ii) if the sum of the signals from the HEMSs indexed by the $\bar{k}$ and $(\bar{k} + 2)$ is correctly decoded by the $(\bar{k} + 1)$ th HEMS and the signal relayed from the $(\bar{k} - 1)$ th HEMS is incorrectly decoded at the $\bar{k}$ th HEMS. Note that, if both the $(\bar{k} + 1)$ th and the $\bar{k}$ th HEMSs incorrectly decode the signals, then, for the BPSK signal, the error at the second stage will cancel the error from the first stage. $\square$

After the second time slot, the HEMSs will receive the sum of the signals forwarded by the adjacent HEMSs. The signal transmission between HEMSs will repeat until the k th HEMS receives the signals from the $(k + 1)$ th HEMS and the ℓ th HEMS from the $(\ell - 1)$ th HEMS. As a result, the chances of error at a certain HEMS can be computed based on the error rates at the HEMSs from whom the particular HEMS has received messages. The following lemma presents the associated mathematical formulation.

Lemma 3. *The probability of incorrectly decoding the ℓ th HEMSs' message at the k th HEMS is approximated as follows:*

$$\Pr_{k,\ell,odd} \approx \sum_{a=1}^{|\ell-k|} \Pr_{k+a-1,k+a} \prod_{b=1}^{\ell-k,b \neq a} (1 - \Pr_{k+b-1,k+b}) \quad (12)$$

where $\Pr_{k+a-1,k+a}$ denotes the probability that the $(k + a - 1)$ th HEMS incorrectly decodes the signal from the $(k + a)$ th HEMS. On the other hand, the second term $(1 - \Pr_{k+b-1,k+b})$ indicates the probability that the $(k + b - 1)$ th HEMS has correctly decoded the signal from the $(k + b)$ th HEMS.

Proof. The chances of error at the k th HEMS for incorrectly estimating the ℓ th HEMS's message will be a function of the probabilities of error at the $|\ell - k|$ HEMSs in between the two HEMSs. At a certain time, there can be J number of HEMSs, where $J \in [1, |\ell - k|]$, $J \in \mathbb{Z}$, from whom the k th HEMS received information. If $J = 1$, the error probability will be quite large given that it is highly probable for one of the HEMSs to decode incorrectly. On the other hand, for $J > 1$, the error probability would be smaller given that it is less likely for more than one HEMSs to incorrectly decode, particularly in high SNR. That is, as SNR increases, it can be considered that the probability of error at the k th HEMS will be dictated by the chances of error at the k th HEMS for decoding a single HEMS (indexed by a) between the k th and the ℓ th HEMS, while the messages from the other HEMSs between k and ℓ are correctly decoded. As a result, the error performance at k th HEMS for obtaining the information from the ℓ th HEMS is presented in (12). $\square$

4.2. k Odd, ℓ Even

In this case, the chances of error for the $\bar{\ell}$ th HEMS to incorrectly decode the message at the first time slot is given as in (9). During time slot 2, the chances of error at the $\bar{k}$ th HEMS is given by (9), with $\bar{\ell}$ replaced by $\bar{k}$. In time slot 3, the $\bar{k}$ th HEMS receives the relayed message from the $\bar{\ell}$ th HEMS, and so, the error probability can be written as in (10). Similarly, in the fourth time slot, the error probability at the $\bar{\ell}$ th HEMS is given by (10) with $\bar{k}$ replaced by $\bar{\ell}$.

The process continues until the $(\frac{|\ell-k|+1}{2} + 1)$ th time slot. Then, similar to Lemma 3, the probability that the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS incorrectly decodes the k th HEMS's message is as follows:

$$P_{k, \frac{|\ell-k|+1}{2}} \approx \sum_{a=1}^{\frac{|\ell-k|+1}{2}-k} P_{k+a-1, k+a} \prod_{b=1, b \neq a}^{\frac{|\ell-k|+1}{2}-k} (1 - P_{k+b-1, k+b}) \quad (13)$$

Similarly, the probability that the $(\frac{|\ell-k|+1}{2})$ th HEMS incorrectly decodes the $(\ell - 1)$ th HEMS's signal can be represented as in (13), with k replaced by $\ell - 1$. The error probability for the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS decoding the $(k + 1)$ th and ℓ th HEMSs' message can be obtained in a similar manner. After the $(\frac{|\ell-k|+1}{2} + 2)$ th time slot, the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS receives the sum of the signal from the $(\frac{|\ell-k|+1}{2})$ th HEMS. Thus, the chances that the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS incorrectly estimates the message from the k th HEMS can be given as in the following lemma.

Lemma 4. The probability that the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS makes an error in estimating the k th HEMS's message in the $(\frac{|\ell-k|+1}{2} + 2)$ th time slot is given by the following:

$$\begin{aligned} Pr_{\frac{|\ell-k|+1}{2}+1, k} &\approx Pr_{\frac{|\ell-k|+1}{2}+1, \frac{|\ell-k|+1}{2}} (1 - Pr_{\frac{|\ell-k|+1}{2}, k}) (1 - Pr_{\frac{|\ell-k|+1}{2}, k-1}) (1 - Pr_{\frac{|\ell-k|+1}{2}+1, \ell-1}) \\ &+ (1 - Pr_{\frac{|\ell-k|+1}{2}+1, \frac{|\ell-k|+1}{2}}) Pr_{\frac{|\ell-k|+1}{2}, i} (1 - Pr_{\frac{|\ell-k|+1}{2}, \ell-1}) (1 - Pr_{\frac{|\ell-k|+1}{2}+1, \ell-1}) \\ &+ (1 - Pr_{\frac{|\ell-k|+1}{2}+1, \frac{|\ell-k|+1}{2}}) (1 - Pr_{\frac{|\ell-k|+1}{2}, k}) Pr_{\frac{|\ell-k|+1}{2}, \ell-1} (1 - Pr_{\frac{|\ell-k|+1}{2}+1, \ell-1}) \\ &+ (1 - Pr_{\frac{|\ell-k|+1}{2}+1, \frac{|\ell-k|+1}{2}}) (1 - Pr_{\frac{|\ell-k|+1}{2}, k}) (1 - Pr_{\frac{|\ell-k|+1}{2}, \ell-1}) Pr_{\frac{|\ell-k|+1}{2}+1, \ell-1} \end{aligned} \quad (14)$$

Here, $Pr_{\frac{|\ell-k|+1}{2}+1, \frac{|\ell-k|+1}{2}}$ denotes the chances of error at the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS for decoding the $(\frac{|\ell-k|+1}{2})$ th HEMS's message. The term $Pr_{\frac{|\ell-k|+1}{2}+1, \ell-1}$ denotes the probability that the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS makes an error about the $(\ell - 1)$ th HEMS's message.

Proof. The $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS receives the $\hat{V}_{k,\ell-1}$ from the $(\frac{|\ell-k|+1}{2})$ th HEMS. Then, it utilises the $(\ell - 1)$ th HEMS's signal (which it has decoded earlier on) to extract the signal from the k th HEMS. The probability that the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS makes an error in estimating the k th HEMS's message can be described using the following scenarios. These include the following: (i) the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS makes an error in estimating the $(\frac{|\ell-k|+1}{2})$ th HEMS's message, (ii) the $(\frac{|\ell-k|+1}{2})$ th HEMS obtains an incorrect estimate of the k th HEMS's message, (iii) the $(\ell - 1)$ th HEMS's message is estimated inaccurately by the $(\frac{|\ell-k|+1}{2})$ th HEMS, and (iv) the $(\ell - 1)$ th HEMS's information is inaccurately obtained by the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS. Since BPSK signals are considered, the $(\frac{|\ell-k|+1}{2} + 1)$ th HEMS incorrectly decodes the signal from the k th HEMS if any one or any three of these events happen at the same time. If two or four events take place at the same time, the error will be automatically corrected for the BPSK signals. At a high SNR, the probability of having one of the four events is significantly larger than the probability of having three of the four events at the same time. As a result, the chances of error for the $(\frac{|\ell-k|+1}{2})$ th HEMS's estimation of the k th HEMS's message is expressed using (14). $\square$

Similarly, the chances of error at the $(\frac{|\ell-k|+1}{2})$ th HEMS about the message from the ℓ th HEMS can be represented using (14), with the term $\frac{|\ell-k|+1}{2} + 1$ replaced by $\frac{|\ell-k|+1}{2}$, and vice versa. Moreover, the terms k and $\ell - 1$ are replaced by $k + 1$ and ℓ , respectively. After the $(\frac{|\ell-k|+1}{2} + 3)$ th time slot, the $(\frac{|\ell-k|+1}{2} + 1)$ th and the $(\frac{|\ell-k|+1}{2})$ th HEMSs forward the signals from the k th and the ℓ th HEMSs to the nearby HEMSs until the k th and the ℓ th HEMS receives the signal from each other. Then, using Lemma 3 and 4, the error performance at the k th HEMS for the erroneous detection of the ℓ th HEMS's signal is given by the following:

$$\begin{aligned} \Pr_{k,\ell,\text{even}} \approx & \sum_{a=1}^{\frac{|\ell-k|+1}{2}+1-k} \Pr_{k+a-1,k+a} \prod_{b=1}^{\frac{|\ell-k|+1}{2}+1-k, b \neq a} (1 - \Pr_{k+b-1,k+b}) (1 - \Pr_{\frac{|\ell-k|+1}{2}+1,k}) \\ & + \prod_{a=1}^{\frac{|\ell-k|+1}{2}+1-k} (1 - \Pr_{k+a-1,k+a}) \Pr_{\frac{|\ell-k|+1}{2}+1,k} \end{aligned} \quad (15)$$

Similarly, the error performance at the ℓ th HEMS when detecting the k th HEMS's messages can be written using (15), with k replaced by ℓ and vice versa. Now, if there are K number of HEMSs in total, then, there will be $K + 1$ cases where an odd-indexed HEMS will decode the signal of an even-indexed HEMS and $K - 1$ cases where an even-indexed HEMS will decode the signal of an odd-indexed HEMS when K is odd. On the other hand, if K is even, in $K - 2$ cases, an odd-indexed HEMS will decode the signal of an even-indexed HEMS, and in K cases, an even-indexed HEMS will decode the signal of an odd-indexed HEMS. As a result, the average error probability for decoding a HEMS's signal in the microgrid can be represented as follows:

$$\Pr_{avg} = \begin{cases} \frac{1}{4K} (2(K + 1)\Pr_{k,\ell,\text{odd}} + 2(K - 1)\Pr_{k,\ell,\text{even}}) & K=\text{odd} \\ \frac{1}{4K-4} (2(K - 2)\Pr_{k,\ell,\text{odd}} + 2K\Pr_{k,\ell,\text{even}}) & K=\text{even} \end{cases}$$

Note that the error performance at the HEMSs for the scenarios when k is even and ℓ is odd can be written in a similar manner as in (14). On the other hand, the error performance at the k th and ℓ th HEMSs for the scenarios when both k and ℓ are even will be the same as that for the scenarios when both k and ℓ are odd.

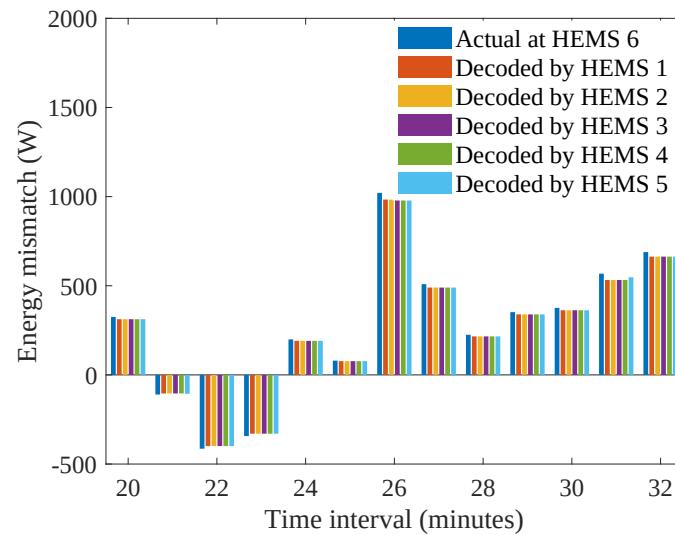
5. Numerical Simulation Results

In this section, a microgrid with six HEMSs is considered. These HEMSs are associated with houses connected to solar panels and the HEMSs can communicate about the surplus generation or energy deficit between each other as well as the central controller. The energy generation and consumption data are obtained from [54]. The data are collected for a particular day in July over 48 intervals, each of which is 30 min long. The nearby HEMSs have a channel gain of 100 between them, unless otherwise specified. The error probabilities are computed for 100 time frames, and then, the results are averaged.

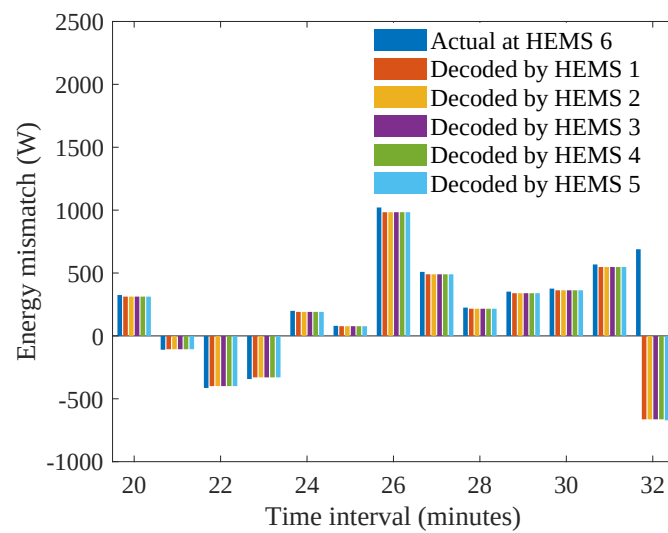
The system under simulation operates in the following manner. HEMSs 1, 3 and 5 transmit their messages, whereas HEMSs 2 and 4 receive the sum of signals and HEMS 6 receives the signal from HEMS 5. Then, HEMS 2, 4 and 6 transmit their signal, HEMSs 3 and 5 receive the sum of the signals and HEMS 1 receives the signal from HEMS 2. Then, HEMS 2 forwards the sum of signals from HEMSs 1 and 3 to them, and as a result, HEMSs 1 and 3 receive each other's messages. Similarly, HEMS 4 forwards the sum of signals and HEMSs 3 and 5 receive each other's signals. HEMS 5 also obtains the message from HEMS 6. After this, HEMSs 3 and 5 forward the sum of the signals received at the beginning, HEMS 2 obtains the signal from HEMS 3 and HEMS 4 obtains the signal from HEMS 5. HEMS 2 also receives the signal from HEMS 1. Then, HEMS 3 computes the sum of HEMS 1 and 5's signals, HEMS 5 computes the sum of HEMSs 3 and 6 and forwards that. Eventually, HEMS 2 receives the message from HEMS 5, HEMS 4 receives the signal from HEMS 1, HEMS 3 from HEMS 6 and HEMS 6 from HEMS 3. The process continues until all HEMSs obtain the signals from every other HEMS.

Figure 2a–c show the actual and decoded energy mismatch information about HEMS 6 at all the other HEMSs for three different schemes. The first one is with the proposed scheme, the second one is with direct DF and the third one is with the central control centre. The energy mismatch value is first calculated at each HEMS by subtracting the generation value from the demand value. Thus, a positive energy mismatch value corresponds to excess demand, whereas the negative energy mismatch value corresponds to excess generation. The results show the energy mismatch information decoded for an SNR level of 5 dB. From the aforementioned figures, it can be seen that the proposed scheme performs at an equivalent level compared to the two other schemes, even at a low SNR. As identified in Remarks 1 and 2, the proposed scheme requires a smaller number of time slots to extract the messages compared to the two other schemes. For example, HEMS 6 can be decoded by all HEMSs by time slot 6 in the proposed scheme, whereas this is possible only after time slot 30 with direct DF and after time slot 12 with a central control centre. Thus, the proposed scheme can significantly reduce the latency while maintaining comparable reliability.

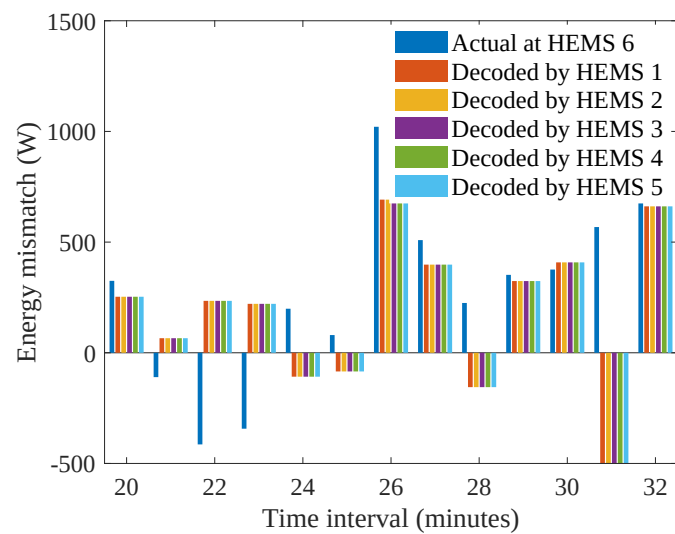
In the following part of this section, we illustrate the error probability results rather than the decoded energy mismatch information for the sake of conciseness and clarity.



(a) Proposed scheme



(b) Direct decode-and-forward



(c) Through central control centre

Figure 2. Energy mismatch information at HEMS 6 decoded by all other HEMSs when the proposed scheme, direct decode-and-forward and transmission through central control centre are used.

5.1. Error Performance at Different HEMSs

Figure 3a–d illustrate the comparison of the proposed scheme and the direct DF scheme for the error probability at all the HEMSs when they decode the energy mismatch information of HEMS 6. The results are demonstrated for the following two different cases: when the channel gains for all channel coefficients are the same (Figure 3a,b) and when the channel gain for the channel between HEMS 1 and 2 drops to 50 (Figure 3c,d). From Figure 3a,b, it is clear that the error probability at the HEMSs is greater in the proposed scheme in comparison with the error probability when direct DF is considered and all channels are in good conditions. It can be seen that the BER is the highest at HEMS 1 for both schemes, since the message from HEMS 6 has to pass through all the intermediate HEMSs. The SNR penalties used to achieve the same error performance as the direct decode–and–forward scheme are 2 dB, 2 dB, 4 dB, 5 dB and 2 dB at HEMS 1, 2, 3, 4 and 5, respectively. This indicates that the error performance at the middle HEMSs is more significantly affected, as compared to the other HEMSs, when the proposed scheme is considered. This represents the worst case error performance for the proposed scheme for a given channel gain condition.

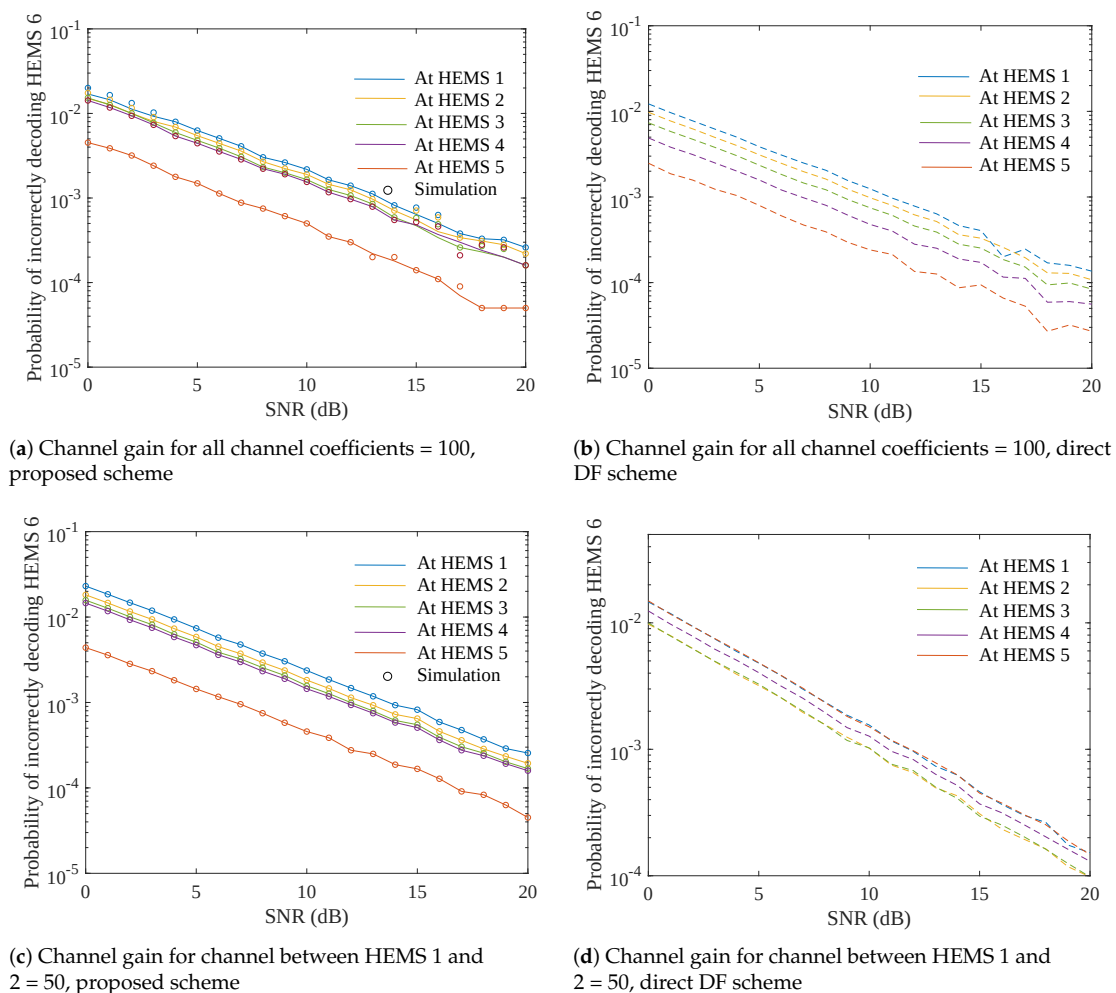


Figure 3. Error performance comparison for the proposed scheme and direct decode–and–forward when HEMS 6 is decoded by all other HEMSs under different channel conditions.

Figure 3c,d show the error performance when the channel gain for channels between HEMS 1 and 2 drops to 50. From the results, we can obtain that the error performance at HEMS 1 for direct DF worsens significantly. In addition, the error performance at HEMS 5 with direct DF degrades more compared to the error performance with the

proposed scheme. The reason is that the error accumulates at each stage when direct DF is used. On the other hand, with the proposed scheme, some HEMSs will be able to decode the message correctly, even if some of the other HEMSs suffer poor channel conditions. The SNR penalties used to achieve the same error performance as the direct decode-and-forward scheme are 2 dB, 2 dB, 1.8 dB and 0.8 dB at HEMSs 1, 2, 3 and 4, respectively. Thus, the difference in error performance at the middle HEMSs becomes smaller when the channel gains are not equal and the channel conditions are poor during the first stage. Note that it is possible to develop an adaptive strategy that switches between direct DF and the proposed communication based on the scenarios when the channel conditions are equally good for the HEMSs and when channel conditions are worse for some HEMSs. However, this requires a detailed analysis of the thresholds used for switching between communication schemes and is beyond the scope of this paper.

5.2. Impact of Channel Gains

In this section, the impact of channel gains between different HEMSs is illustrated for the error probabilities at individual HEMSs and the average error performance. Figure 4 shows the BER sensitivity index at HEMS 1 and 6, when the channel gain between HEMSs 3 and 4 dropped from 100 to 50 for a number of different channel gains between other HEMSs. The sensitivity index is computed using the ratio of the change in BER and the change in the channel gain. For example, if a 50% change in channel gain results into a 5% change in BER, then, the sensitivity index will be 0.1. A negative sensitivity index indicates that the BER with $v_{34} = 50$ is low compared to the case when all channel gains are set as equal. Thus, it can be seen that the sensitivity index is negative when all other channel gains are less than 50, and then, it becomes positive. That is, when the other channel gains are more than 50, dropping v_{34} to 50 worsens the error performance. It can be seen that the sensitivity index increases with the channel gain between other HEMSs. So, as the channel gain increases, the impact of changing v_{34} increases. However, with 20 dB SNR, the impact becomes less strong after a certain channel gain. Also, it can be observed that with low SNR, the sensitivity is high when the channel gain of the other HEMSs is low, and it becomes less strong when the channel gains are high. On the other hand, when the SNR is high, the sensitivity becomes low, with a smaller channel gain for the other HEMSs, and it increases when the channel gain is high.

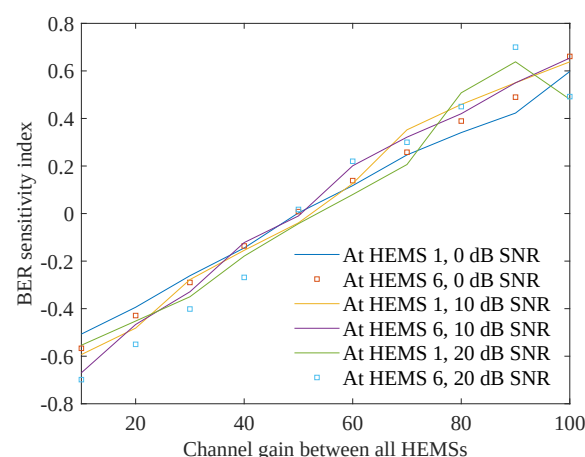


Figure 4. BER sensitivity at HEMS 1 and 6 with different channel conditions and different SNR.

Figure 5 shows the the average BER of the overall microgrid and the relative impact of the channel gain between different HEMSs. The channel gain in one of the links that have been set to 50 while the other channel gains remain 100. It can be seen that the drop in channel gain between HEMSs 1 and 2, as well as HEMSs 5 and 6, does not significantly

degrade the average BER performance. As the channel gain drops for intermediate links such as the one between HEMSs 2 and 3 or HEMSs 4 and 5, there is a moderate increase in average BER. The channel between the middle HEMSs (HEMSs 3 and 4) has a greater influence on the average BER, which is similar to the worst case error probability results, as shown in Figure 3a. Note that there is a constant SNR penalty between the different scenarios, and the average BER decreases at the same rate with an increase in the SNR. There is a 2 dB SNR penalty between the best possible scenario (when the channel gain degrades for the HEMSs at the edge) and worst possible scenario (when the channel gain degrades for the HEMSs in the middle).

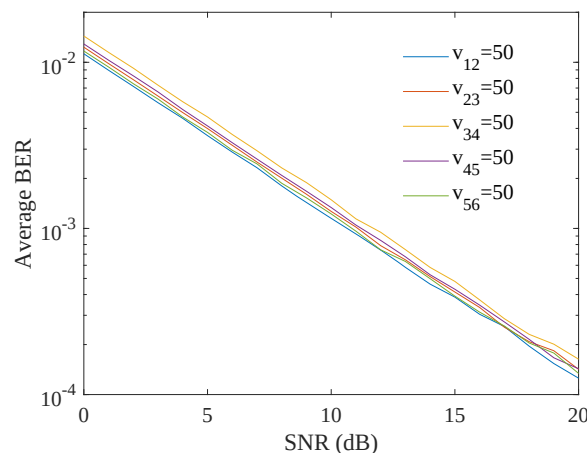


Figure 5. Average BER performance of the overall network with different channel gains.

6. Conclusions

We have proposed and evaluated a new communication protocol to enable the direct communication between IoT-enabled HEMSs in a smart microgrid for decentralised energy trading applications. For the proposed protocol, the mathematical expressions for error performance at a HEMS as well as the error performance of the overall network have been derived analytically. The simulation results have illustrated that the error performance of the proposed scheme depends more on the channel conditions between the HEMSs in the middle of the network. It has also been observed that the proposed scheme suffers a certain SNR penalty in order to achieve the same error probabilities as by direct DF, which depends on which HEMS is decoding the message. The SNR penalty between the two schemes becomes smaller with poorer channel conditions, indicating that the proposed scheme offers a much more efficient mechanism rather than the direct DF without a significant SNR penalty. Despite the benefits offered by the proposed communication scheme, it is vulnerable to a number of issues, including channel estimation error, interference and security risks. To address them in the future, the detailed mathematical modelling of channel estimation errors will be investigated, and advanced coding techniques will be explored to eliminate the impact of such errors. In addition, beamforming and precoding techniques will be integrated to nullify the impact of interferences in the unlicensed spectrum. Physical layer security-based approaches such as integrating artificial noise may prevent the leakage of private information from prosumers. Future studies will also consider an adaptive strategy that switches between communication schemes involving the control centre, direct DF and the proposed scheme based on variations in the channel conditions at the HEMSs and control centre.

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